Exercise 2 – Ultrasound transmission and reception

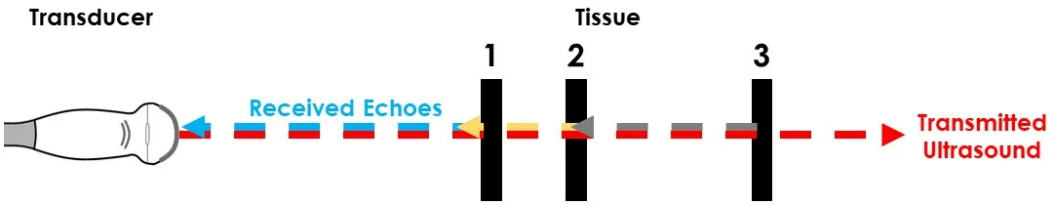
Main learning aims:

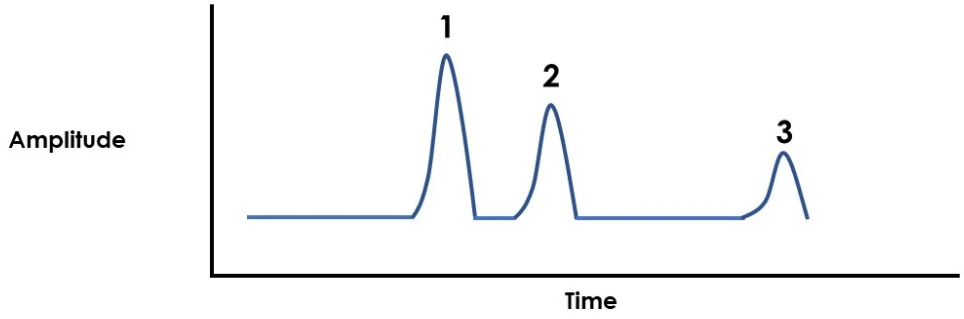
PART 1:

- To learn about transmit pulses in ultrasound imaging, here Gaussian-shaped and square waves
- To learn about the impulse and frequency response of an ultrasound transducer
- To learn about how the transducer transfer function influences the physically transmitted pulse

PART 2:

- To learn about how we filter the received signal to improve SNR
- To learn about complex (IQ) demodulation to represent our received signal in baseband
- To learn how to combine the simplified signal model to generate an ultrasound A-mode image, i.e. a single line generated from the envelope of the received signal





Let's go!

PART 1: TRANSMISSION

The ultrasound transmitter come as different types, where the most complex can transmit arbitrarily shaped pulses (*arbitrary transmit pulser,* based on a digital-to-analog converter), while the most common is a simpler *bipolar square-wave pulser* which transmits a rectangular pulse train we call a *square wave*.

Let's first generate and plot these pulses, see the text box below for suggestions to parameters and tips on functions that can be used.

1. Generate a Gaussian-weighted sinusoidal pulse, a "Gaussian pulse"
    a. NB: Use a time vector from -T/2 to T/2 to get the full pulse from start to finish (you will see)
    b. Calculate and plot the frequency spectrum of the pulse

2. Generate a square wave pulse with a given frequency f0 and number of oscillations / length in time
    a. Center the pulse around zero similarly as the Gaussian pulse and plot in the same subplot to compare
    b. Calculate and plot the frequency spectrum of this pulse.

    It is nice to plot the signals and frequency spectra in the same subplots to compare.

3. What key differences do you observe between the two pulses in the frequency domain?

4. Measure (or calculate analytically) the spatial pulse length (SPL) of the Gaussian pulse by measuring the full width half maximum of the signal envelope, report it in [mm]. What is the radial resolution?

Tips:

- Suggestions for initial parameters:
  - fs = 250MHz, sampling frequency of the transmitter
  - f0 = 2.5MHz, initial pulse frequency
  - bw = 0.3, fractional bandwidth (i.e. bw_abs = f0*bw)
  - T = 10 microseconds, observation time (make time vector from -T/2 to T/2)

- The Gaussian-weighted sinusoidal pulse can be generated using the *gauspuls* function in Matlab or *gausspuls* in Scipy

- The square wave pulse can be generated using the *square* function in Matlab or Scipy.
  - The length of the pulse can initially be set to the inverse of the bandwidth defined for the Gaussian pulse, then they will have a similar effective length.
    - T_p=1/f0, one period of pulse
    - N_p = int(1/bw), number of periods
  - You can then pad the output of the square with zeros to get it to the same length in time as for the Gaussian pulse for easy plotting.

- To estimate the power spectrum you can use the *Periodogram approach* that we used in Exercise 1 based on the FFT. Remember to set the x and y-limits to visualize the relevant part of the spectrum, e.g. [-15, 15] MHz, and [-45, 5] dB for a normalized spectrum.

- To calculate the pulse length using the full width half maximum (FWHM):
  - Estimate the pulse envelope using the Hilbert function: sig_env = abs(hilbert(env))
  - Find the points where the envelope amplitude has dropped to half the maximum amplitude on each side and find the length in time. Convert to space using a speed of sound of 1540 m/s.

Transducer impulse / frequency response

The transducer is a resonant electromechanical system, which only supports transmitting and receiving ultrasound in a certain band of frequencies. This is termed the *transducer bandwidth*. On the other hand, a transducer *Q-factor* defines how resonant the transducer is at its main resonance frequency (center frequency), and the higher the Q-factor the lower the bandwidth of the transducer, but also the more sensitive the transducer is to receiving ultrasound at its resonant frequency. So, choices and trade-offs are made when designing a transducer. For imaging, bandwidth is very important, and we are usually able to achieve fractional bandwidths of 80-90% at lower frequencies (e.g. cardiac imaging). For higher frequencies, e.g. > 10MHz, the fractional bandwidth achieved is lower due to practical manufacturing / mechanical reasons (e.g. 60-70%). The transducer transfer function dictates how the transmitted pulse is filtered from an electrical signal to an acoustic pulse.

Let's first define a simplified transducer transfer function and plot its frequency response. A simple transducer transfer function is to define the transducer impulse response as a Gaussian weighted sinusoid, like the transmit pulse defined above.

5. Make the transducer impulse response as a Gaussian-weighted sinusoid. As a start, use a transducer center frequency f_xd=2.5MHz, and fractional bandwidth bw_xd = 0.4.
6. Plot the transducer impulse response and the frequency response in separate subplots.
7. Plot the transmit signal(s) in the same subplots to compare.

Influence on transmitted pulse

The transducer acts as a band pass filter on the transmitted pulse. This filtering operation can be represented by a convolution in time with the transducer impulse response, or multiplication in frequency with its frequency response. We can here use Numpy *convolve* or Matlab *conv* function to filter the transmitted signal with the transducer impulse response.

8. Filter the transmitted pulses by convolving them with the transducer impulse response as the filter defined in 5).

9. Plot the filtered pulse and corresponding power spectrum in different subplots. Add the original pulse and its power spectrum to compare. **Note**: You can normalize the

output of the convolution to better compare the output shape to the original pulse (so that both have an output amplitude of 1).

10. What do you observe for:
    a. The Gaussian pulse after the transducer?
    b. The square wave pulse after the transducer?

11. Change the transmit pulse frequency to 4MHz (up from 2.5MHz). What do you observe now?

12. What do you think happens to the pulse energy that is outside the transducer bandwidth?

PART 2: RECEPTION

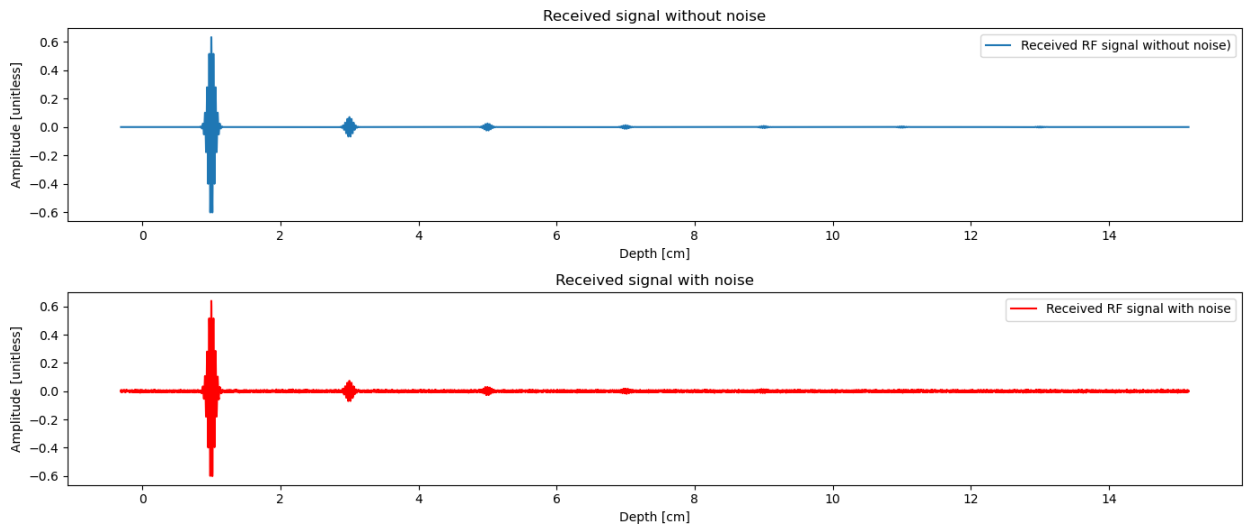
Let's make a simplified simulation of the received pulse, by representing the propagation of ultrasound as (yet) another filter in time, which represents a delay (propagation time) and attenuation of the propagating pulse. This can be modeled by what is called the Green's function:

$$G(r,t) = \frac{\delta(t-\frac{r}{c})}{4\pi r},$$

where r is the depth of the scatterer and c is the speed of sound. The nominator represents an impulse at time t=r/c when the scatterer is met, and the denominator represents how the amplitude diminishes with distance for a point source. To model the two-way response, i.e. transmit *and* receive propagation, we can simply *square this equation*. We will finally add noise to the received signal. The simulation thus results in a received pulse which is a delayed and scaled copy of the transmitted pulse, with added noise.

We will emulate a cardiac imaging setup, and add a number of virtual reflecting targets at 1 – 15 cm. You can assume the noise level to be constant, while the reflected signal intensity changes in depth according to the Greens function (**note:** The attenuation mentioned here is not frequency-dependent attenuation).

1. Start by creating an array of scatterer positions, e.g. [1, 3, 5, 7, 9, 11, 13] cm
2. Define a depth axis where you will but in the impulses from the Green function, and scale the impulse with depth as given by the formula.
    a. Use the pulse echo response, meaning that you square the Green's function.
3. Create the reflected signal by convolving the *transmitted pulse* (after the transducer) with the propagation filter defined above, using varying positions *r* (e.g. between 1 and 15 cm) of the scatterer.
    a. The pulse we defined may have zeros before and after, this will cause a delay and extra computations in the convolution. Therefore:
        i. Cut the pulse to limit the amount of trailing zeros.
        ii. Also cut the time axis for the transmitted pulse similarly.
    b. After the convolution operation the time axis will be stretched according to the length of the pulse and receive depth/time axis, and we thus need redefine it using, pseudocode: t_conv = linspace(t_tx[0]+t_rx[0],t_tx[-1]+t_rx[-1],len(s_rx)).
4. Add noise to the received signal, the results should look something like this:



Where the bottom plot is the received signal with noise.

Receive filtering

5. Create a receive filter that passes the pulse but attenuates the noise outside the pass band. You can use a similar code as you did in exercise 1. You can use either a FIR and IIR filter as you like.
6. Filter the received signal and compare the output with the signal before filtering (plot results).

Depth-dependent amplification / Time-gain compensation (TGC)

To reduce the dynamic range and show the reflected signals we compensate for the depth dependent attenuation. This is usually referred to as time-gain compensation (TGC), where time and depth is related through t = 2*r/c.

7. Create a depth-gain profile based on your knowledge of the signal model above and how the signal reduces with depth. Multiply this gain profile with the received (and filtered) signal.

8. Plot the results before and after TGC.

IQ demodulation

Let's make the baseband representation of the signal to: 1) reduce the sampling frequency, and 2) prepare for further envelope detection and A-mode display.

9. Use the Hilbert transform to generate the analytical signal
10. Calculate the power spectrum of the signal before and after the Hilbert transform.
    a. What do you see?
11. Demodulate the signal after Hilbert transformation down to baseband.
    a. Multiply (downmix) with a complex exponential at the center frequency of the pulse: *sig_iq = sig_hilbert*e(-i*2*pi*f0*t)*

Envelope detection and A-mode

The ultrasound A-mode imaging modality was the original one used in the 50s and 60s, and is based on plotting the envelope of the received RF-signal (echoes). Let's end this lecture by making our first A-mode.

1. You can find the envelope of the signal as the absolute value of the IQ-signal that you created in the previous part.
2. Make a plot of the A-mode line in [dB]
3. Celebrate!

Congratulations, you have made your first ultrasound A-mode line!